

Companion to Bird & de Moor, Algebra of Programming

§8. Thinning Algorithms

§8.1. Thinning

(8.1)

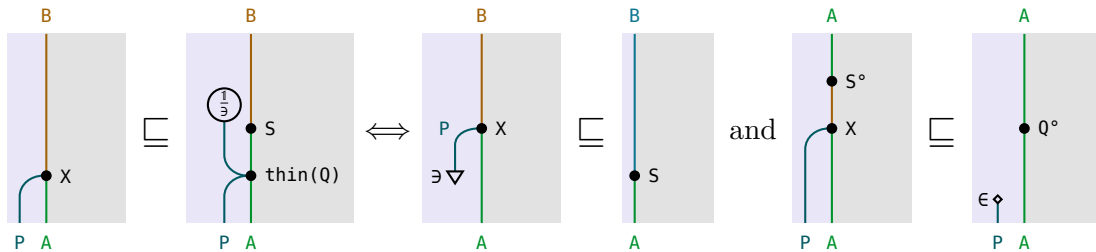
$\text{thin}(Q) \triangleq \sup_n (\epsilon \setminus (Q^\circ \epsilon))$ — $\text{thin}(Q)$ takes a set ys to a subset xs that keeps a Q -lower bound of every $a \in ys$.
 $\text{thin}(\leq) \ ys \ \text{thin}(\leq) \ xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \leq a).$

(8.1a)

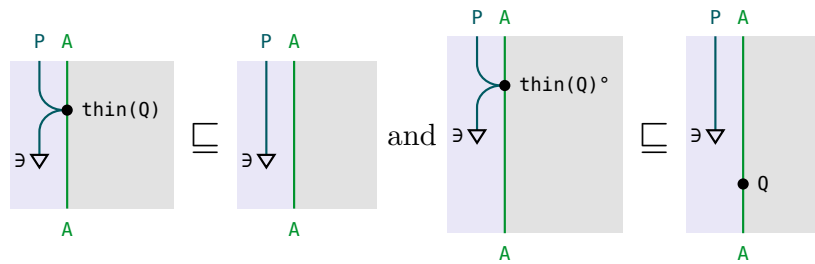
the law	what it says
1 $X \sqsubseteq \frac{S}{\sup} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \epsilon$	S is the whole algebra: it produces every candidate that $\text{thin}(Q)$ thins. Everything kept is an S -value, and every S -value has a Q -lower bound among the kept ones
2 $\text{thin}(Q) \ni \sqsubseteq \ni$	everything a thinning keeps was in the set
3 $\text{thin}(Q)^\circ \ni \sqsubseteq \ni Q$	every element of the set has a Q -lower bound among the kept ones
4 $Q \sqsubseteq R \implies \text{thin}(Q) \sqsubseteq \text{thin}(R)$	the fewer pairs Q relates, the fewer subsets count as thinnings
5 $1 \sqsubseteq Q \implies 1 \sqsubseteq \text{thin}(Q)$ $Q \sqsubseteq Q \implies \text{thin}(Q) \text{thin}(Q) \sqsubseteq \text{thin}(Q)$	keeping everything is always a legal thinning
6 $Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge R R \sqsubseteq R \implies \text{est}(R) = \text{thin}(Q) \text{est}(R)$ $1 \sqsubseteq Q$ $\implies \text{est}(R) \sqsubseteq \text{thin}(Q) \text{est}(R)$	thin-introduction: thinning first cannot lose an R -minimum
7 $\text{thin}(Q) \supseteq \text{est}(Q) \frac{1}{\sup} \quad (8.2) \text{ — } (8.2) \text{est}(Q) \frac{1}{\sup} \sqsubseteq \text{thin}(Q)$	thin-elimination: keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on
8 $\frac{S}{\sup} \text{thin}(Q) \supseteq \frac{S}{\sup} \text{est}(R) \frac{1}{\sup}$ (8.3), $Rn(S^\circ S) \sqsubseteq Q$ — thin-elimination variant	the usable variant: R need only refine Q between values S gives one argument
9 $P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q) \quad (8.4) \text{ — } (8.4)$ $P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q)$	thinning each member set is a thinning of the union

§8.1.1. $X \sqsubseteq \frac{S}{\sup} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \epsilon$ — the definition of $\frac{S}{\sup} \text{thin}(Q)$

(8.1.1a)



(8.1.1b)



$X \sqsubseteq \frac{S}{\sup} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \epsilon$ at $X \triangleq \text{thin}(Q)$, $S \triangleq \ni$: the left side is $\text{thin}(Q) \sqsubseteq \text{thin}(Q)$, since $\frac{\ni}{\sup} = \wedge(\ni) = 1$

§8.1.2. $Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge R R \sqsubseteq R \implies \text{est}(R) = \text{thin}(Q) \text{est}(R)$ — thinning first, then taking the smallest, is the same as taking the smallest

$$1 \sqsubseteq Q \Rightarrow \text{est}(R) \sqsubseteq \text{thin}(Q) \text{ est}(R) \quad (8.1.2a)$$

the $\sqsubseteq$ half: keeping everything is a thinning — $1 \sqsubseteq Q$



1	$\text{est}(R)$	$1 \sqsubseteq \text{thin}(Q) \text{ — } 1 \sqsubseteq Q$ $\Rightarrow 1 \sqsubseteq \text{thin}(Q), Q$ $Q \sqsubseteq Q$ $\Rightarrow \text{thin}(Q) \text{ thin}(Q)$ $\sqsubseteq \text{thin}(Q)$
2	$\sqsubseteq \text{thin}(Q) \text{ est}(R)$	

$$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists \quad (8.1.2b)$$

the $\sqsubseteq$ half, first condition of the UP of est at $X \triangleq \text{thin}(Q) \text{ est}(R) \text{ — } X \sqsubseteq \text{est}(R) \Leftrightarrow X \sqsubseteq \exists \wedge \in X \sqsubseteq R^\circ$



1	$\text{thin}(Q) \text{ est}(R)$	$R \cap S \sqsubseteq R$
2	$\sqsubseteq \text{thin}(Q) \exists$	$\text{thin}(Q) \exists \sqsubseteq \exists$
3	$\sqsubseteq \exists$	

$$Q \sqsubseteq R \wedge R R \sqsubseteq R \Rightarrow \in \text{thin}(Q) \text{ est}(R) \sqsubseteq R^\circ \quad (8.1.2c)$$

the $\sqsubseteq$ half, second condition



1	$\in \text{thin}(Q) \text{ est}(R)$	$\in \text{thin}(Q) \sqsubseteq Q^\circ \in$
2	$\sqsubseteq Q^\circ \in \text{est}(R)$	$\in \text{est}(R) \sqsubseteq R^\circ$
3	$\sqsubseteq Q^\circ R^\circ$	$Q \sqsubseteq R$
4	$\sqsubseteq R^\circ R^\circ$	$R R \sqsubseteq R$ $\Rightarrow R^\circ R^\circ \sqsubseteq R^\circ$
5	$\sqsubseteq R^\circ$	

§8.1.3. Properties of thinning

$$\mathbf{est}(Q) \stackrel{1}{\exists} \sqsubseteq \mathbf{thin}(Q)$$

(8.2)

the singleton holding a Q -least member of a set is a thinning of that set

$$\mathbf{est}(Q) \stackrel{1}{\exists} \ni \sqsubseteq \ni$$

every member of the singleton is a member of the set — $\triangleq \mathbf{thin}, \ni/\ni$ half



1	$\mathbf{est}(Q) \stackrel{1}{\exists} \ni$	$\stackrel{1}{\exists} \ni = 1$
2	$= \mathbf{est}(Q)$	$R \cap S \sqsubseteq R$
3	$\sqsubseteq \ni$	

$$\in \mathbf{est}(Q) \stackrel{1}{\exists} \sqsubseteq Q^\circ \in$$

the singleton's member stands in Q to every member of the set — $\triangleq \mathbf{thin}, \in/(Q^\circ \in)$ half

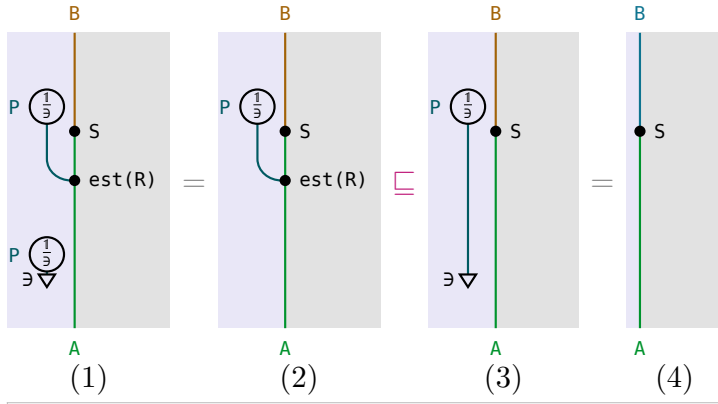


1	$\in \mathbf{est}(Q) \stackrel{1}{\exists}$	$\in \mathbf{est}(R) \sqsubseteq R^\circ$
2	$\sqsubseteq Q^\circ \stackrel{1}{\exists}$	$\stackrel{1}{\exists} \sqsubseteq \in$
3	$\sqsubseteq Q^\circ \in$	

$$R \cap (S^\circ S) \subseteq Q \Rightarrow \frac{S}{S} \text{est}(R) \frac{1}{S} \subseteq \frac{S}{S} \text{thin}(Q) \quad (8.3)$$

$$\frac{S}{S} \text{est}(R) \frac{1}{S} \ni \in S$$

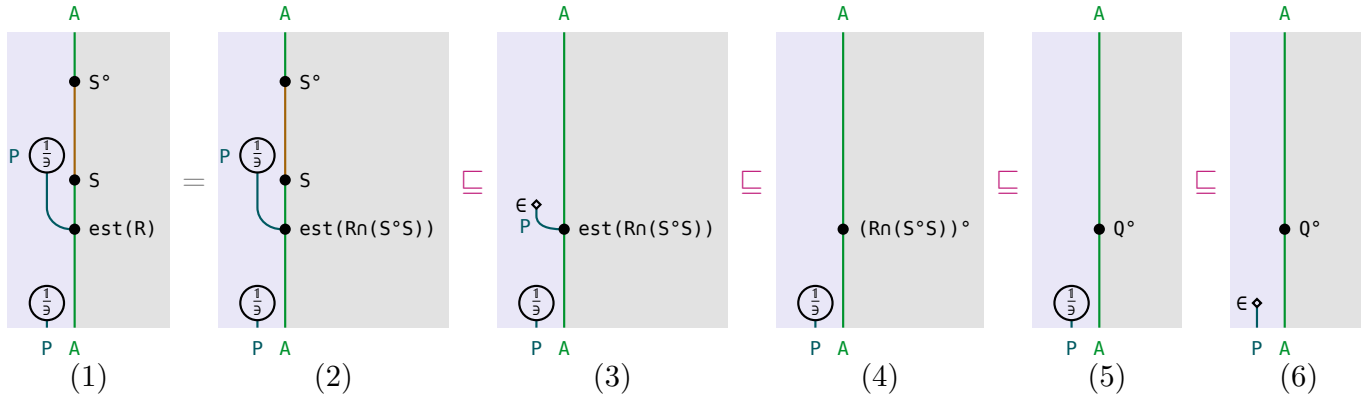
every member of the singleton is a value S returns — $X \subseteq \frac{S}{S} \text{thin}(Q) \Leftrightarrow X \ni \in S \wedge S^\circ X \subseteq Q^\circ \in$, first condition



1	$\frac{S}{S} \text{est}(R) \frac{1}{S} \ni$	$\frac{1}{S} \ni = 1$
2	$= \frac{S}{S} \text{est}(R)$	$R \cap S \subseteq R$
3	$\subseteq \frac{S}{S} \ni$	$\frac{R}{S} \ni = R$
4	$= S$	

$$R \cap (S^\circ S) \subseteq Q \Rightarrow S^\circ \frac{S}{S} \text{est}(R) \frac{1}{S} \subseteq Q^\circ \in$$

the singleton's member stands in Q to every value S returns at the same argument — $X \subseteq \frac{S}{S} \text{thin}(Q) \Leftrightarrow X \ni \in S \wedge S^\circ X \subseteq Q^\circ \in$, second condition



1	$S^\circ \frac{S}{S} \text{est}(R) \frac{1}{S}$	$\frac{S}{S} \text{est}(R \cap (S^\circ S)) = \frac{S}{S} \text{est}(R)$
2	$= S^\circ \frac{S}{S} \text{est}(R \cap (S^\circ S)) \frac{1}{S}$	$T^\circ \frac{1}{S} \subseteq \in$
3	$\subseteq \in \text{est}(R \cap (S^\circ S)) \frac{1}{S}$	$\in \text{est}(R) \subseteq R^\circ$
4	$\subseteq (R \cap (S^\circ S))^\circ \frac{1}{S}$	$R \cap (S^\circ S) \subseteq Q$
5	$\subseteq Q^\circ \frac{1}{S}$	$\frac{1}{S} \subseteq \in$
6	$\subseteq Q^\circ \in$	

$$\text{preorder}(Q) \wedge S \text{ Pres } Q \Rightarrow (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \sqsubseteq (\frac{S}{\exists} \text{thin}(Q))$$

Theorem 8.1

thinning at every step of the reduce is a thinning of the whole candidate set

$$(\frac{F(\exists) S}{\exists} \text{thin}(Q)) \ni \sqsubseteq (S)$$

everything the thinning fold keeps is a value of the plain fold — $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$, first condition, by fusion from the chain below



1	$\frac{F(\exists) S}{\exists} \text{thin}(Q) \ni$	
2	$= \frac{F(\exists) S}{\exists} \text{thin}(Q) \ni$	$\text{thin}(Q) \ni \sqsubseteq \exists$
3	$\sqsubseteq \frac{F(\exists) S}{\exists} \exists$	$\frac{R}{\exists} \exists = R$
4	$= F(\exists) S$	

$$\text{preorder}(Q) \wedge S \text{ Pres } Q \Rightarrow (S)^\circ (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \sqsubseteq Q^\circ \in$$

every value of the plain fold is Q -above one the thinning fold keeps — $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$, second condition, by hylomorphism theorem from the chain below



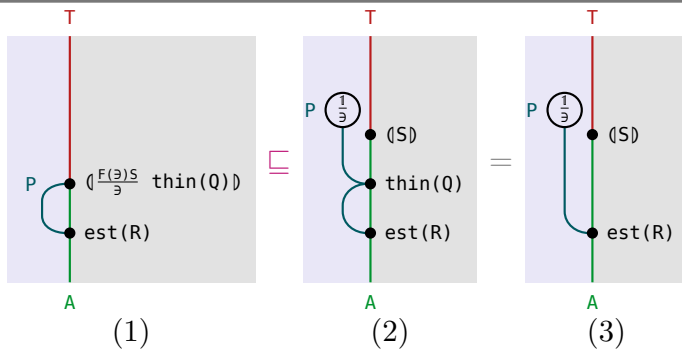
1	$S^\circ F(Q^\circ \in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	
2	$= S^\circ F(Q^\circ) F(\epsilon) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	$S \text{ Pres } Q$
3	$\sqsubseteq Q^\circ S^\circ F(\epsilon) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	$\implies S^\circ F(Q^\circ) \sqsubseteq Q^\circ S^\circ$
4	$\sqsubseteq Q^\circ \in \text{thin}(Q)$	$S^\circ F(\epsilon) \frac{F(\exists) S}{\exists} \sqsubseteq \epsilon$
5	$\sqsubseteq Q^\circ Q^\circ \in$	$\epsilon \text{thin}(Q) \sqsubseteq Q^\circ \in$
6	$\sqsubseteq Q^\circ \in$	

(8.1.3a)

step	type	example	what it does
S:F(A)→A is 8.1's algebra; example F(X)=V+V×X, S=[wrap,cons], candidate a=cons(v,p)			
S°	A→FA	cons(v,p) ↦ r(v,p)	takes a apart into the input of its last step
F(Q°∈)	FA→F(PA)	r(v,p) ↦ r(v,ps), some p'∈ps with p' Q p	replaces each recursive component by a set holding a Q-no-worse one; v stays
$\frac{F(\exists)S}{\exists}$	F(PA)→PA	r(v,ps) ↦ {cons(v,t) t∈ps}	picks one element from each set, combines by S, collects every result
thin(Q)	PA→PA	{cons(v,t) t∈ps} ↦ ys	keeps a subset; every dropped element is Q-beaten by a kept one
$\sqsubseteq Q^\circ \in$	A→PA	ys holds y with y Q cons(v,p') Q cons(v,p) = a	the thinned set still holds an element Q-no-worse than a (S monotonic on Q, Q transitive)

$$Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge S \text{ Pres } Q \Rightarrow \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \text{est}(R) \sqsubseteq \frac{(S)}{\exists} \text{est}(R) \quad \text{Corollary 8.1}$$

the thinning fold refines the optimisation problem itself



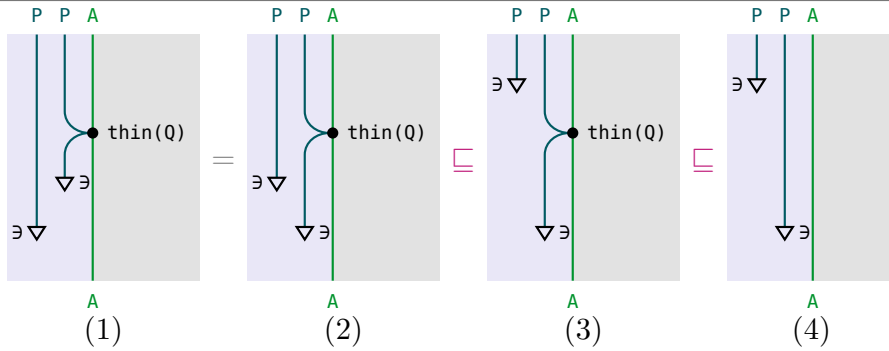
1	$\langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \text{est}(R)$	Theorem 8.1 $\text{preorder}(Q) \wedge S \text{ Pres } Q$
2	$\sqsubseteq \frac{(S)}{\exists} \text{thin}(Q) \text{est}(R)$	$\Rightarrow \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \frac{(S)}{\exists} \text{thin}(Q)$
3	$= \frac{(S)}{\exists} \text{est}(R)$	

$$P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q) \quad (8.4)$$

thinning each member set, then taking the union, is a thinning of the union

$$P(\text{thin}(Q)) \text{ union} \exists \sqsubseteq \exists$$

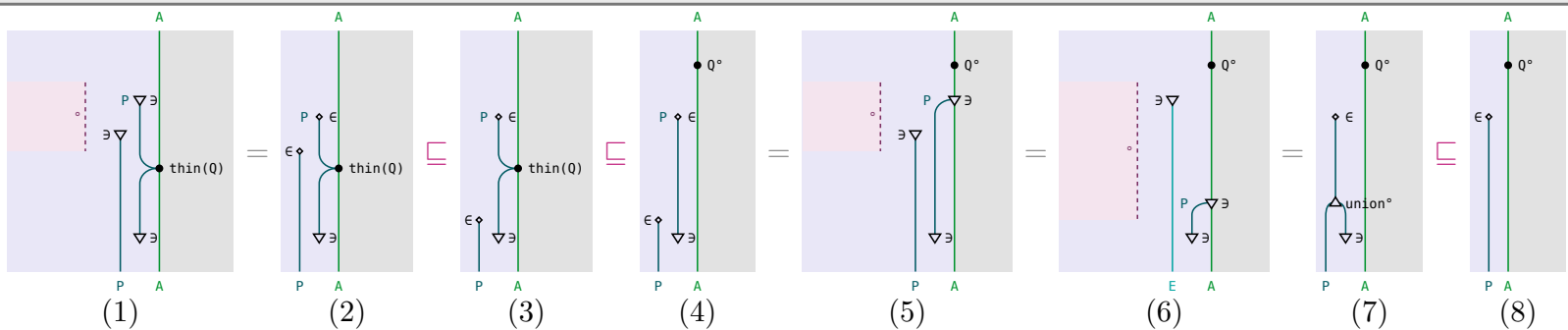
every member of the result is a member of a member set — $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \exists \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$, first condition



1	$P(\text{thin}(Q)) \text{ union} \exists$	$\text{union} \exists = \exists \exists$
2	$= P(\text{thin}(Q)) \exists \exists$	$P(R) \exists \sqsubseteq \exists R$
3	$\sqsubseteq \exists \text{thin}(Q) \exists$	$\text{thin}(Q) \exists \sqsubseteq \exists$
4	$\sqsubseteq \exists \exists$	

$$(\exists \exists)^\circ P(\text{thin}(Q)) \text{ union} \sqsubseteq Q^\circ \in$$

every member of a member set has a Q -lower bound in the result — $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \exists \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$, second condition



1	$(\exists \exists)^\circ P(\text{thin}(Q)) \text{ union}$	$(R S)^\circ = S^\circ R^\circ$
2	$= \in \in P(\text{thin}(Q)) \text{ union}$	$\in P(R) \sqsubseteq R \in$
3	$\sqsubseteq \in \text{thin}(Q) \in \text{union}$	$\in \text{thin}(Q) \sqsubseteq Q^\circ \in$
4	$\sqsubseteq Q^\circ \in \in \text{union}$	$(R S)^\circ = S^\circ R^\circ$
5	$= Q^\circ (\exists \exists)^\circ \text{union}$	$\text{union} \exists = \exists \exists$
6	$= Q^\circ (\text{union} \exists)^\circ \text{union}$	$(R S)^\circ = S^\circ R^\circ$
7	$= Q^\circ \in \text{union}^\circ \text{union}$	$\text{union} = E(\exists)$
8	$\sqsubseteq Q^\circ \in$	

§8.2. Paths in a layered network

(8.2a)

	name	type	definition	meaning
1	F	$\mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$	$F(B, X) = B + (B \times X)$	a path is either one vertex a , or a vertex a followed by a path xs
2	α	$F(A, LA) \rightarrow LA$	$\alpha = [\text{wrap}, \text{cons}]$	build a path: wrap one vertex as $[a]$, or put a in front of xs
3	wrapz	$A \rightarrow LA \times \mathbb{N}$	$\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle, a \mapsto ([a], 0)$	a path of one vertex, no edge, cost 0
4	consw	$A \times (LA \times \mathbb{N}) \rightarrow LA \times \mathbb{N}$	$(a, (xs, n)) \mapsto ([a] \# xs, \text{wt}(a, \text{head}(xs)) + n)$	put a in front of the path; the cost adds the weight of the new edge $a \rightarrow \text{head}(xs)$
5	cost	$LA \rightarrow \mathbb{N}$	$\text{cost} \triangleq \langle [\text{wrapz}, \text{consw}] \rangle \pi_2$	the cost of $[a_0, \dots, a_n]$ is $\text{wt}(a_0, a_1) + \dots + \text{wt}(a_{n-1}, a_n)$
6	R	$LA \rightarrow LA$	$R \triangleq \text{cost} \leq \text{cost}^\circ$	xs is cheaper than ys
7	$L(R)$	$LA \rightarrow LB$	$\langle F(\exists, 1) \alpha \rangle = L(\exists)$	$L(\exists)$ relates a list of layers to every path that chooses one vertex from each layer
8	minpath	$L(PA) \rightarrow LA$	$\text{minpath} \sqsubseteq \frac{L(\exists)}{\exists} \text{est}(R)$	the input $[x_0, \dots, x_n] : L(PA)$ is a list of layers, each a set of vertices; a path chooses one vertex from each layer, and minpath returns a cheapest one
9	Q	$LA \rightarrow LA$	$Q \triangleq R \cap (\text{head} \text{ head}^\circ)$	like R , and the two paths also start at the same vertex: $\text{head}(xs) = \text{head}(ys)$
10	l	$B \rightarrow B + C$	$a \mapsto l(a)$	mark a value as the left alternative
11	r	$C \rightarrow B + C$	$b \mapsto r(b)$	mark a value as the right alternative
12	cpl	$PB \times A \rightarrow P(B \times A)$	$\text{cpl} \triangleq \text{cp}(1 \times A)$	pair every element of the set with the value beside it
13	cpr	$A \times PB \rightarrow P(A \times B)$	$\text{cpr} \triangleq \text{cp}(A \times 1)$	pair the value beside the set with every element of the set
14	$\frac{F(\exists, 1)}{\exists}$	$F(PB, X) \rightarrow PF(B, X)$	$\frac{F(\exists, 1)}{\exists} = [P(l), \text{cpl } P(r)]$	cpl transpose: from layer x and a set of paths ps , choose a vertex $a \in x$
15	$\frac{F(1, \exists)}{\exists}$	$F(B, PX) \rightarrow PF(B, X)$	$\frac{F(1, \exists)}{\exists} = [\frac{1}{\exists} P(l), \text{cpr } P(r)]$	cpr transpose: from a vertex a and a set of paths ps , choose a tail $xs \in ps$
16	step	$A \times P(LA) \rightarrow LA$	$\text{step} \triangleq \text{cpr } P(\text{cons}) \text{est}(R)$	put a in front of every path in ps and keep a cheapest one

(8.2b)

the law	what it says
1 $F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$	$F(\exists, 1) \alpha$ is monotonic on Q ; on R it is not, since the next edge can cost arbitrarily much
2 $S \text{ head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$	S head is simple, which gives $R n (S^\circ S) \sqsubseteq Q$: between two paths S builds from one argument, equal cost and equal head already means Q

§8.2.1. Derivation

$\frac{F(\exists, 1) \quad P(\frac{F(1, \exists) \alpha}{\exists} \text{ est}(R))}{\exists} \text{ est}(R) \sqsubseteq \frac{F(\exists, 1) \alpha}{\exists} \text{ est}(R) \quad (8.2.1a)$ a least-cost path in a layered network, as a fold over the layers	
- Why it is needed: $\text{est}(R)$ keeps one cheapest w of the set $\Lambda(S)(x)$; $\text{thin}(Q)$ needs every dropped z to be Q-beaten by a kept one. For $\{w\}$ alone to be a thinning we need $w Q z$ for every $z \in \Lambda(S)(x)$. - We have $w R z$ (w is cheapest) and $w (S^\circ S) z$ (both are built by S from the same argument x); the hypothesis turns the two into $w Q z$: among candidates from one argument, cheaper already means Q-better.	
(1) (2) (3)	
1 $\frac{F(\exists, 1) \quad P(\frac{F(1, \exists) \alpha}{\exists} \text{ est}(R))}{\exists} \text{ est}(R)$	$\frac{F(\exists, 1) \quad P(\frac{F(1, \exists) \alpha}{\exists} \text{ est}(R))}{\exists} \sqsubseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{ thin}(Q)$
2 $\sqsubseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{ thin}(Q) \text{ est}(R)$	$\frac{F(\exists, \exists) \alpha}{\exists} \text{ thin}(Q) \text{ est}(R) \sqsubseteq \frac{F(\exists, 1) \alpha}{\exists} \text{ est}(R)$
3 $\sqsubseteq \frac{F(\exists, 1) \alpha}{\exists} \text{ est}(R)$	

$S \triangleq F(1, \exists) \alpha, R n (S^\circ S) \sqsubseteq Q \Rightarrow \frac{F(\exists, 1) \quad P(\frac{F(1, \exists) \alpha}{\exists} \text{ est}(R))}{\exists} \sqsubseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{ thin}(Q) \quad (8.2.1b)$ thinning the algebra of a layered network costs no more than taking the program's two cases	
(1) (2) (3) (4) (5) (6) (7)	
1 $\frac{F(\exists, 1) \quad P(\frac{F(1, \exists) \alpha}{\exists} \text{ est}(R))}{\exists}$	
2 $= \frac{F(\exists, 1) \quad P(\frac{F(1, \exists) \alpha}{\exists} \text{ est}(R))}{\exists}$	$P(\frac{1}{\exists}) E(\exists) = 1$
3 $= \frac{F(\exists, 1) \quad P(\frac{F(1, \exists) \alpha}{\exists} \text{ est}(R) \frac{1}{\exists})}{\exists} E(\exists)$	(8.3) $R n (S^\circ S) \sqsubseteq Q$
4 $\sqsubseteq \frac{F(\exists, 1) \quad P(\frac{F(1, \exists) \alpha}{\exists}) \quad P(\text{thin}(Q)) \quad E(\exists)}{\exists}$	$\Rightarrow \frac{S}{\exists} \text{ est}(R) \frac{1}{\exists} \sqsubseteq \frac{S}{\exists} \text{ thin}(Q)$
5 $\sqsubseteq \frac{F(\exists, 1) \quad P(\frac{F(1, \exists) \alpha}{\exists}) \quad E(\exists) \quad \text{thin}(Q)}{\exists}$	(8.4) $P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q)$
6 $= \frac{F(\exists, 1) \quad F(1, \exists) \quad \alpha}{\exists} \text{ thin}(Q)$	$\bigvee \frac{S}{\exists} = \bigvee \frac{P(\frac{S}{\exists})}{\exists} \text{ union}$
7 $= \frac{F(\exists, \exists) \alpha}{\exists} \text{ thin}(Q)$	$F(R, 1) F(1, S) = F(R, S)$
$= \text{cp}(F) P(\alpha) \text{ est}(R) = [\text{wrap}, \text{step}]$ at the layered network $(F(A, X) = A + A \times X, B \text{ the paths } V^+, \alpha = [\text{wrap}, \text{cons}]): \frac{F(1, \exists)}{\exists} P(\alpha) \text{ est}(R) = [\text{wrap}, \text{step}] - \text{step} \triangleq \text{cpr}$ $P(\text{cons}) \text{ est}(R)$ $= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} - \frac{F(\exists, 1)}{\exists} = [P(l), \text{cpl } P(r)]$	

$$\begin{array}{ccccc}
 F(A, X) & = & A & + & A \times X \\
 & & | & & | \quad | \\
 & & R & & R \quad S \\
 & & \downarrow & & \downarrow \quad \downarrow \\
 F(A', X') & = & A' & + & A' \times X'
 \end{array}$$

(8.2.1c)

u $F(R, S) v$ holds iff one of:

- u, v are both in l and $d R d'$: the leaf moves by R ;

- u, v are both inr and $p_1 \ R \ q_1 \wedge p_2 \ S \ q_2$: the vertex moves by R , the rest by S .
- An inl never relates to an inr : $R+S \triangleq [R \ \text{inl}, S \ \text{inr}]$ returns to the summand it came from, and $F(1)=1$ allows no crossing.
- $F(\exists, \exists)$ relates $\text{inr}(\{a, b\}, \{p, q\})$ to $\text{inr}(a, p), \text{inr}(a, q), \text{inr}(b, p), \text{inr}(b, q)$, and $\text{inl}(\{a, b\})$ to $\text{inl}(a), \text{inl}(b)$.

§8.3. Implementing thin

(8.3a)

	name	type	definition	meaning
1	setify	$[A] \rightarrow PA$	$(\text{setify } \exists)(xs, x) \iff x \in xs$	the set of a list's elements
2	cup	$PA \times PA \rightarrow PA$	$\text{cup} \triangleq \frac{(\pi_1 \exists) \cup (\pi_2 \exists)}{\exists}$	the union of two sets
3	cp(F)	$F(PA) \rightarrow PFA$	$\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$	every F-shaped value whose parts are chosen from the given sets
4	listcp(F,A,X)	$\text{relator}(\text{list}(X)) \rightarrow \text{list}(\text{relator}(X))$	$\text{listcpFn}(\text{zer}, x) = []$, $\text{listcpFn}(\text{PolyC.one}, u) = [u, ()]$, $\text{listcpFn}(\text{const}(a), c) = [c]$, $\text{listcpFn}(\text{arg}_1, a) = [a]$, $\text{listcpFn}(\text{arg}_2, xs) = xs$, $\text{listcpFn}(l \oplus a, l(u)) =$ $\text{list}(\text{inl}, \text{listcpFn}(l, u))$, $\text{listcpFn}(a \oplus r, r(v)) =$ $\text{list}(\text{inr}, \text{listcpFn}(r, v))$, $\text{listcpFn}(l \otimes r, (u, v)) =$ $\text{cppFn}(\text{listcpFn}(l, u), \text{listcpFn}(r, v))$	by the shape of F: a constant or arg_1 is a one-element list, arg_2 the list itself, a sum lists its summand, a product pairs every element of one list with every element of the other
5	linear	$P(\text{PolyC}(\iota))$	$\text{linear}(\text{zer}) = \text{true}$, $\text{linear}(\text{PolyC.one}) = \text{true}$, $\text{linear}(\text{const}(a)) = \text{true}$, $\text{linear}(\text{arg}_1) = \text{true}$, $\text{linear}(\text{arg}_2) = \text{true}$, $\text{linear}(l \oplus r) = \text{linear}(l) \wedge$ $\text{linear}(r)$, $\text{linear}(l \otimes r) = \text{linear}(l) \wedge \text{linear}(r) \wedge (\neg$ $\text{hasArg}_2(l) \vee \neg \text{hasArg}_2(r))$	no product in F has the argument on both sides
6	sort($\leq$)	$PA \rightarrow LA$	$\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ordered}(\leq)$	read a set back as one of its $\leq$ -ordered listings
7	bump(Q)	$A \times [A] \rightarrow [A]$	$(a, []) \mapsto [a]$, $\text{bump}(Q)((a, [b] \# xs), ys) \iff Q(a, b) \wedge ys =$ $[a] \# xs \vee Q(b, a) \wedge ys = [b] \# xs \vee \neg Q(a, b) \wedge \neg Q(b, a) \wedge ys =$ $[a] \# [b] \# xs$	put a in front of a thinned list, dropping whichever of a and the old head the other beats
8	thinlist(Q)	$[A] \rightarrow [A]$	$\text{thinlist}(Q) \triangleq (\text{setify } \text{thin}(Q) \text{ setify}^\circ) \text{ n subseq}$	only drop elements, and on the set of the elements be a thinning by Q
9			$T \sqsubseteq \text{thinlist}(Q) \iff T \sqsubseteq \text{subseq} \wedge T \text{ setify} \sqsubseteq \text{setify } \text{thin}(Q)$	what a list implementation of $\text{thin}(Q)$ must do: only drop elements, and thin the set
10			$\text{preorder}(Q) \implies ([\text{nil}, \text{bump}(Q)]) \sqsubseteq \text{thinlist}(Q)$	an implementation: thin a list in one pass, bumping each element into the thinned rest
11	minlist(Q)	$[A] \rightarrow A$	$\text{minlist}(Q) \triangleq \text{setify } \text{est}(Q)$	a Q-least element of the list

§8.3.1. Sorting sets

(8.3.1a)

the law	what it says
1 $\text{preorder}(Q) \wedge \text{connected}(Q) \Rightarrow$ $\text{bumpFold}(Q)([a]\#xs,ys) \Leftrightarrow (\exists w. \text{minlist}(Q)([a]\#xs,w)$ $\wedge ys = [w])$ (8.5)	what thinning should come to when it can: one element
2 $\text{sort}(\leq) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(\leq)$ (8.6) — (8.6) $\text{sort}(\leq) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(\leq)$	the thinning may run after the sort
3 $\text{sort}(\leq) \text{minlist}(R) \sqsubseteq \text{est}(R)$ (8.7)	a minimum of the sorted list is a minimum of the set
4 $\text{sort}(f \leq f^\circ) \text{list}(f) \sqsubseteq P(f) \text{sort}(\leq)$ (8.8)	shunt a function through a sort
5 $\text{Coreflexive}(p) \Rightarrow \text{sort}(\leq) \text{filter}(p) \sqsubseteq E(p) \text{sort}(\leq)$ (8.9) $p : A \rightarrow A$	filtering a sorted list sorts the restricted set
6 $\text{preorder}(\leq) \wedge \text{connected}(\leq) \Rightarrow (\text{sort}(\leq) \times \text{sort}(\leq))$ $\text{merge}(\leq) \sqsubseteq \text{cup sort}(\leq)$ (8.10)	merging two sorted lists sorts their union
7 $F(\text{sort}(\leq)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sort}(F(\leq))$ (8.11), $FX=L+E \times X$	listcp is the list implementation of the cartesian product $\text{cp}(F)$
8 $\text{linear}(F) \Rightarrow \text{relator}(\text{sort}(\leq)) \text{listcp}(F,A,X) \sqsubseteq$ $\text{cp}(\text{relator}(F,A)) \text{sort}(\text{relator}(\leq))$ (8.11), F polynomial and linear	the same for every linear polynomial F

sort($\leq$) thinlist(Q) $\sqsubseteq$ thin(Q) sort($\leq$)

(8.6)

every list got by sorting the set and then thinning it by Q is also got by thinning the set by Q and then sorting.

1 $\text{sort}(\leq) \text{thinlist}(Q)$	$\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ordered}(\leq)$
2 $\triangleq \text{setify}^\circ \text{ordered thinlist}(Q)$	$\text{ordered thinlist}(Q) \sqsubseteq \text{thinlist}(Q) \text{ordered}$
3 $\sqsubseteq \text{setify}^\circ \text{thinlist}(Q) \text{ordered}$	$\text{setify}^\circ \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{setify}^\circ$
4 $\sqsubseteq \text{thin}(Q) \text{setify}^\circ \text{ordered}$	
5 $= \text{thin}(Q) \text{sort}(\leq)$	

$\text{Coreflexive}(p) \wedge f \text{ Pres} \leq \Rightarrow F(\text{sort}(\leq)) \text{ listcp list}(f) \text{ filter}(p) \sqsubseteq \frac{F(\exists) f p}{\exists} \text{ sort}(\leq)$ Lemma 8.1	
sorting each argument set ($F(\text{sort}(\leq))$), forming every combination of their elements (listcp), applying f to each ($\text{list}(f)$) and keeping those that pass p ($\text{filter}(p)$) gives one of the sorted lists of the set of all $f(u)$ that pass p , with u drawn from the argument sets.	
 (1)	 (2)
 (3)	 (4)
 (5)	 (6)
1 $F(\text{sort}(\leq)) \text{ listcp list}(f) \text{ filter}(p)$	$F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(F(\leq))$
2 $\sqsubseteq \text{cp}(F) \text{ sort}(F(\leq)) \text{ list}(f) \text{ filter}(p)$	$\text{Map}(f)$
3 $\sqsubseteq \text{cp}(F) \text{ sort}(f \leq f^\circ) \text{ list}(f) \text{ filter}(p)$	$\Rightarrow f \text{ Pres } R \Leftrightarrow F(R) \sqsubseteq f R f^\circ$
4 $\sqsubseteq \text{cp}(F) P(f) \text{ sort}(\leq) \text{ filter}(p)$	$\text{sort}(f \leq f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(\leq)$
	$\text{Coreflexive}(p)$
	$\Rightarrow \text{sort}(\leq) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(\leq)$
	$\text{Map}(f)$
	$\Rightarrow P(f) = E(f)$
5 $\sqsubseteq \text{cp}(F) E(f p) \text{ sort}(\leq)$	$\frac{S}{\exists} E(R) = \frac{S R}{\exists}$
6 $= \frac{F(\exists) f p}{\exists} \text{ sort}(\leq)$	

(8.3.1b)

At $F(X)=L+ExX$, $f(r(e,x))=e+x$, $\leq$ is $\leq$, p passes the odd numbers, input $r(3,\{4,1,2\})$.

left side	what it does	result
1 input	an $r(e, xs)$, xs a set	$r(3, \{4, 1, 2\})$
2 $F(\text{sort}(\leq))$	sorts the set in the X place; e is untouched	$r(3, [1, 2, 4])$
3 listcp	pairs e with each element, in list order	$[r(3, 1), r(3, 2), r(3, 4)]$
4 $\text{list}(f)$	f on each	$[4, 5, 7]$
5 $\text{filter}(p)$	keeps the odd ones, order kept	$[5, 7]$

right side	what it does	result
1 input		$r(3, \{4, 1, 2\})$
2 $F(\exists)$	relates $r(3, xs)$ to each $r(3, x)$ with $x \in xs$	$r(3, 4), r(3, 1), r(3, 2)$
3 f		7, 4, 5
4 p	only odd values pass	7, 5
5 $\wedge(\dots)$	collects every result into one set	$\{5, 7\}$
6 $\text{sort}(\leq)$		$[5, 7]$

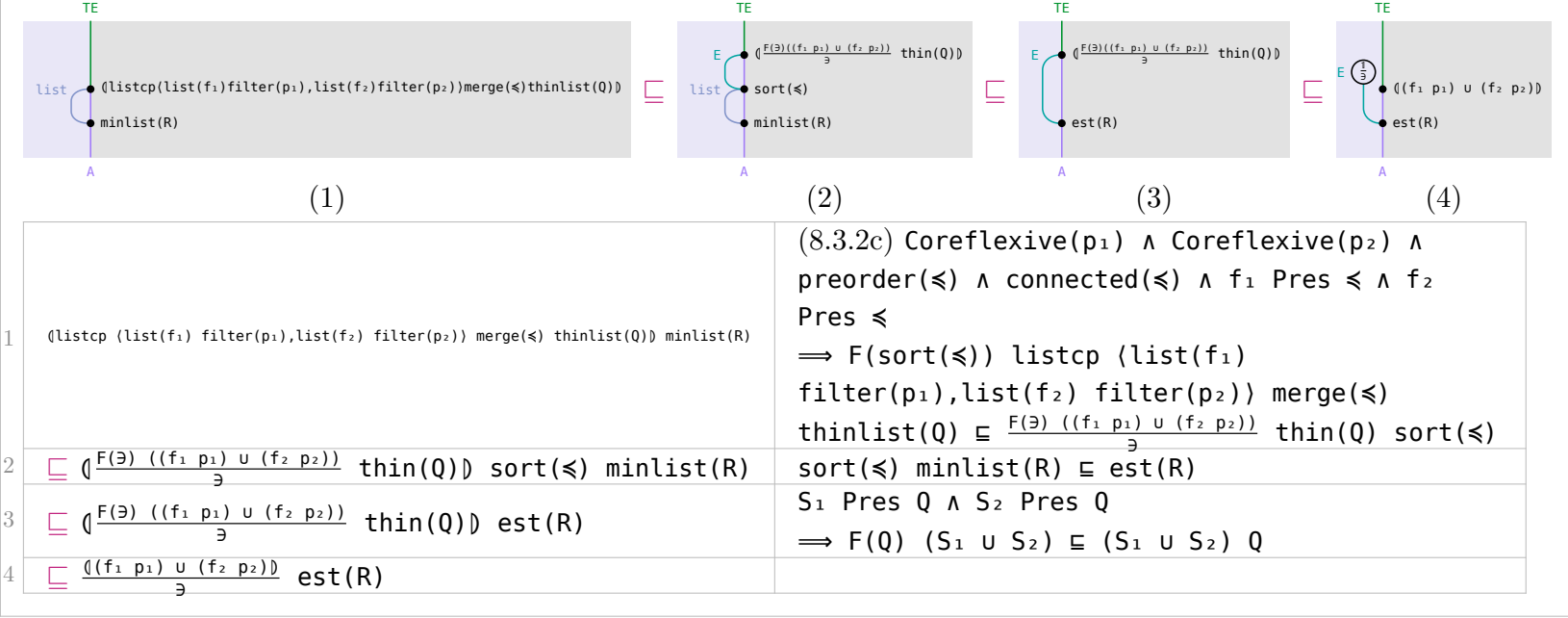
- $\sqsubseteq$: the left's $[5, 7]$ is one result of the right; with ties $\text{sort}(\leq)$ has several orders.
- f preserving $\leq$ is used at $\text{list}(f)$: $[1, 2, 4]$ is sorted, so $[4, 5, 7]$ is; $\text{filter}(p)$ only drops elements. With $f(r(e, x))=e-x$ the left gives $[1, -1]$ after p , not sorted, so $\sqsubseteq$ fails.

§8.3.2. Binary thinning

(8.3.2a)

Binary thinning data: Q a preorder with $Q \sqsubseteq R$ and both $f_1 p_1$, $f_2 p_2$ monotonic on Q ; $\leq$ a connected preorder with both f_1 , f_2 monotonic on $\leq$; R a preorder.

Theorem 8.2
 $\text{Coreflexive}(p_1) \wedge \text{Coreflexive}(p_2) \wedge Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge (f_1 \text{ Pres } \leq \wedge f_2 \text{ Pres } \leq) \implies$
 $(\text{listcp}(\text{list}(f_1) \text{ filter}(p_1), \text{list}(f_2) \text{ filter}(p_2)) \text{ merge}(\leq) \text{ thinlist}(Q)) \text{ minlist}(R) \sqsubseteq$
 $\frac{((f_1 \text{ p}_1) \cup (f_2 \text{ p}_2))}{\exists} \text{ est}(R)$
 a fold on sorted lists of partial solutions, thinned at every step, refines the thinning specification — at binary thinning data.

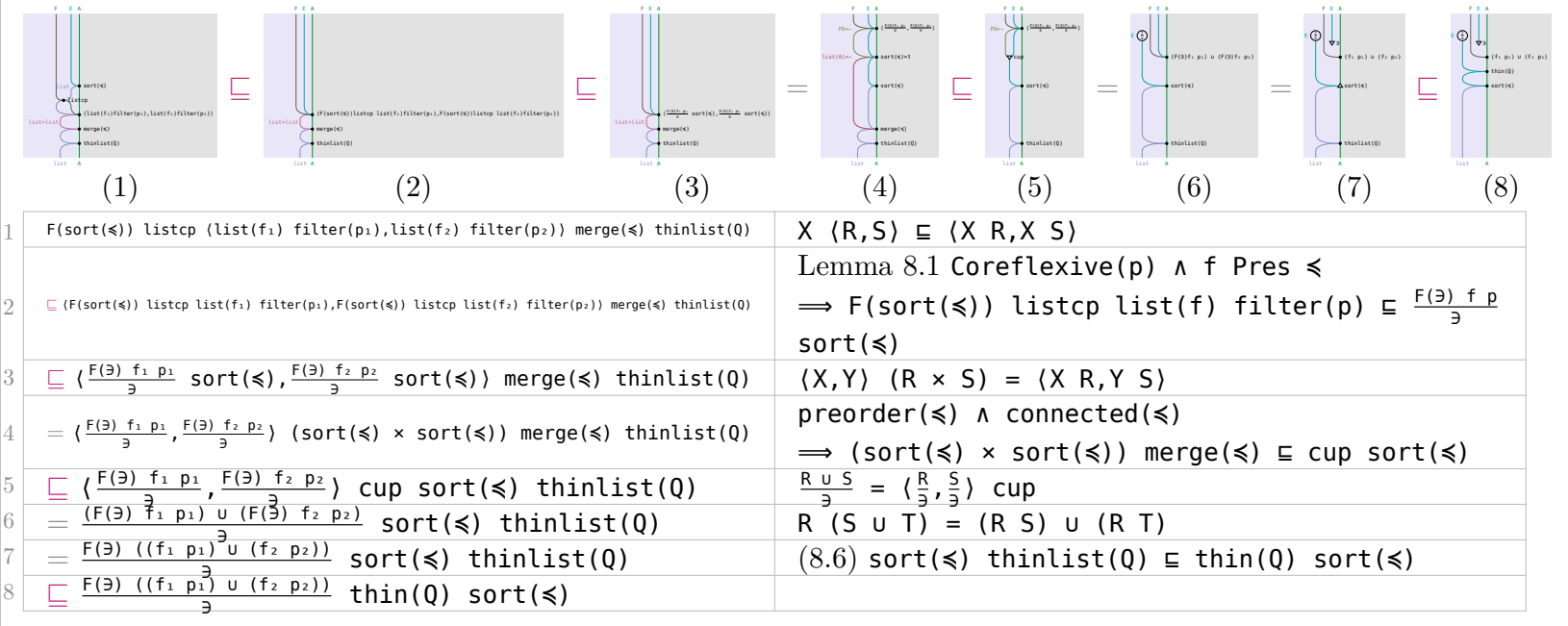


At $F(X)=L+E \times X$, $f_1(\text{inl}(l))=f_2(\text{inl}(l))=0$, $f_1(r(e,x))=x$, $f_2(r(e,x))=e+x$, p_1 passes all, p_2 passes values ≤ 6 , $\leq$ is $\leq$, Q is $=$, R is $\geq$, input $[3,3,1]$. (8.3.2b)

input	$\text{list}(f_1) \text{ filter}(p_1)$	$\text{list}(f_2) \text{ filter}(p_2)$	$\text{merge}(\leq)$	$\text{thinlist}(Q)$
1 $\text{inl}(l)$	$[0]$	$[0]$	$[0,0]$	$[0]$
2 $r(1,[0])$	$[0]$	$[1]$	$[0,1]$	$[0,1]$
3 $r(3,[0,1])$	$[0,1]$	$[3,4]$	$[0,1,3,4]$	$[0,1,3,4]$
4 $r(3,[0,1,3,4])$	$[0,1,3,4]$	$[3,4,6]$	$[0,1,3,3,4,4,6]$	$[0,1,3,4,6]$
5 $\text{minlist}(R)$				6

- $\text{merge}(\leq)$ is one pass: both copies arrive $\leq$ -sorted (f_i monotonic on $\leq$, $\text{filter}(p_i)$ only drops), so it only compares their two heads.
- $\text{thinlist}(Q)$ compares neighbours only; the list is $\leq$ -sorted, so equal values are neighbours and one pass drops every repeat.

Theorem 8.3.2c
 $\text{Coreflexive}(p_1) \wedge \text{Coreflexive}(p_2) \wedge \text{preorder}(\leq) \wedge \text{connected}(\leq) \wedge f_1 \text{ Pres } \leq \wedge f_2 \text{ Pres } \leq \implies$
 $F(\text{sort}(\leq)) \text{ listcp}(\text{list}(f_1) \text{ filter}(p_1), \text{list}(f_2) \text{ filter}(p_2)) \text{ merge}(\leq) \text{ thinlist}(Q) \sqsubseteq$
 $\frac{F(\exists) ((f_1 \text{ p}_1) \cup (f_2 \text{ p}_2))}{\exists} \text{ thin}(Q) \text{ sort}(\leq)$
 sorting the candidate set turns the thinning algebra into an algebra on sorted lists.



§8.4. The knapsack problem

(8.4a)

	name	type	definition	meaning
1	<code>total(f,xs)</code>	<code>Int</code>	$\text{total}(f,xs) \triangleq \text{sum}(\text{list}(f,xs))$, $\text{total } f = \text{list}(f) \text{ sum}$	add up f over the items of the list
2	<code>subseq</code>	<code>[A]→[A]</code>	$xs \text{ subseq } ys \iff \text{subseq}(ys,xs)$	every subsequence of the list
3	<code>[nil,cons]</code>	<code>F[Item]→[Item]</code>	$[\text{nil},\text{cons}] = [\text{nil},\text{cons}]$, $[\text{nil},\text{cons}](\text{l}(u)) = []$, $[\text{nil},\text{cons}](\text{r}(a,xs)) = [a]\#xs$	keep the item: the list constructor
4	<code>[nil,π₂]</code>	<code>Unit + Item × [Item]→[Item]</code>	$[\text{nil},\pi_2](\text{l}(D)) = []$, $[\text{nil},\pi_2](\text{r}(p)) = \pi_2(p)$, $[\text{nil},\pi_2] = [\text{nil},\pi_2]$	drop the item: keep the tail
5	<code>within(w)</code>	<code>[Item]→[Item]</code>	$\text{within}(w)(xs,ys) \iff xs = ys \wedge \text{total}(\text{wt},xs) \leq w$	the packings whose total weight is at most w
6	$([\text{nil},\text{cons}]\text{within}(w)) \cup [\text{nil},\pi_2]$	<code>F[Item]→[Item]</code>	$([\text{nil},\text{cons}]\text{within}(w)) \cup [\text{nil},\pi_2] \triangleq ([\text{nil},\text{cons}]\text{within}(w)) \cup [\text{nil},\pi_2]$, $([\text{nil},\text{cons}]\text{within}(w)) \cup [\text{nil},\pi_2] = [\text{nil},(\text{cons within}(w)) \cup \pi_2]$, $([\text{nil},\text{cons}]\text{within}(w))(u,rs) \iff rs = [\text{nil},\text{cons}](u) \wedge \text{total}(\text{wt},rs) \leq w$	at each item, keep it if the packing still fits, or drop it
7	<code>R</code>	<code>[Item]→[Item]</code>	$R = \text{total vol} \geq \text{total vol}^\circ$, $R(xs,ys) \iff \text{total}(\text{vol},ys) \leq \text{total}(\text{vol},xs)$	x is worth at least as much as y
8	<code>Q(vol,wt)</code>	<code>[Item]→[Item]</code>	$Q(\text{vol},\text{wt}) = R \cap (\text{total wt} \leq \text{total wt}^\circ)$, $Q(\text{vol},\text{wt})(xs,ys) \iff \text{total}(\text{vol},ys) \leq \text{total}(\text{vol},xs) \wedge \text{total}(\text{wt},xs) \leq \text{total}(\text{wt},ys)$	x is worth at least as much as y and weighs no more

(8.4b)

	the law	what it says
1	$(1 \times R) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) R$ FALSE	not monotonic on R: a selection of greater value need not still fit once one more item goes in
2	$F(Q(\text{vol},\text{wt})) [\text{nil},\text{cons}] \text{within}(w) \sqsubseteq [\text{nil},\text{cons}] \text{within}(w) Q(\text{vol},\text{wt})$ $F(Q(\text{vol},\text{wt})) [\text{nil},\pi_2] \sqsubseteq [\text{nil},\pi_2] Q(\text{vol},\text{wt})$,	both halves are monotonic on Q once ties in value are broken by weight
3	$F(R) [\text{nil},\text{cons}] \sqsubseteq [\text{nil},\text{cons}] R$ $F(R) [\text{nil},\pi_2] \sqsubseteq [\text{nil},\pi_2] R$	both algebras are monotonic on R itself, so R is the sort order P

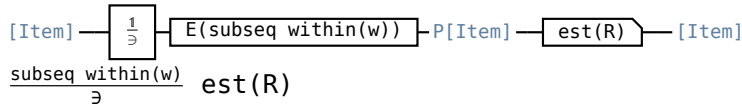
§8.4.1. Derivation

$$0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \Rightarrow \langle \text{listcp } \langle g_1, g_2 \rangle \text{ merge}(R) \text{ thinlist}(Q(\text{vol}, \text{wt})) \rangle \text{ minlist}(R) \sqsubseteq \text{est}(R) \quad (8.4.1a)$$

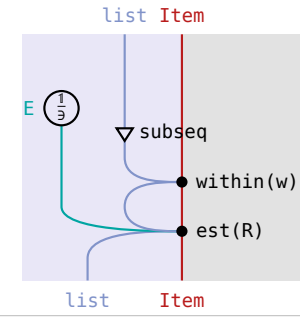
$\frac{\text{subseq within}(w)}{\exists}$

the knapsack problem, as a fold that thins the packings kept at each item

circuit



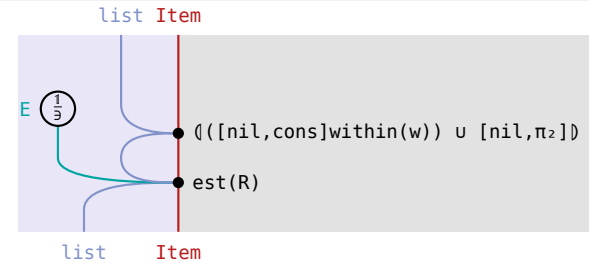
Hinze–Marsden



$$= \langle \langle [\text{nil}, \text{cons}] \text{ within}(w) \rangle \cup [\text{nil}, \pi_2] \rangle \text{ est}(R)$$

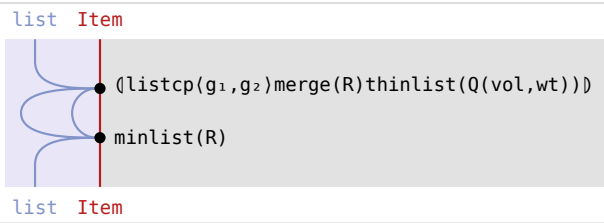
$\frac{\text{subseq within}(w)}{\exists}$

fusion, weights non-negative.



$$\langle \text{listcp } \langle g_1, g_2 \rangle \text{ merge}(R) \text{ thinlist}(Q(\text{vol}, \text{wt})) \rangle \text{ minlist}(R)$$

binary thinning theorem, at $P \triangleq R$, F linear, Q from $F(Q(\text{vol}, \text{wt}))$
 $[\text{nil}, \text{cons}] \text{ within}(w) \sqsubseteq [\text{nil}, \text{cons}] \text{ within}(w) \ Q(\text{vol}, \text{wt})$,
 $F(Q(\text{vol}, \text{wt})) \ [\text{nil}, \pi_2] \sqsubseteq [\text{nil}, \pi_2] \ Q(\text{vol}, \text{wt})$



$$\langle [\text{nil}, \text{cpr } \langle h_1, h_2 \rangle \text{ merge } R \text{ thinlist}(Q)] \rangle \text{ minlist}(R)$$

$\text{listcp} = \text{wrap} + \text{cpr}$, $g_i = [\text{list}(\text{nil}), h_i] - [\text{nil}, \text{cons}] =$
 $[\text{nil}, \text{cons}], [\text{nil}, \pi_2](l(D)) = [], [\text{nil}, \pi_2](r(p)) = \pi_2(p)$;
 $\text{minlist}(R)$ is head, packings coming out in descending value

§8.5. The paragraph problem

(8.5a)

	name	type	definition	meaning
1	Line	Type	$\text{Line} \triangleq \text{L}(\text{Word})$	a line is a non-empty list of words
2	Para	Type	$\text{Para} \triangleq \text{L}(\text{Line})$	a paragraph is a non-empty list of lines
3	$\text{new}(a, \text{xss})$	Para	$\text{new}(a, \text{xss}) \triangleq [[a]] \# \text{xss}$	start a new first line holding only the word a
4	$\text{glue}(a)$	$\text{Para} \rightarrow \text{Para}$	$\text{glue}(a, [\text{xs}]) = [[a] \# \text{xs}]$, $\text{glue}(a, [\text{xs}] \# a') = [[a] \# \text{xs}] \# a'$	put the word a at the front of the first line
5	$[\text{wrap wrap}, \text{new}]$	$\text{F}(\text{Para}) \rightarrow \text{Para}$	$[\text{wrap wrap}, \text{new}](\text{l}(a)) = [[a]]$, $[\text{wrap wrap}, \text{new}](\text{r}(q)) = \text{new}(\pi_1(q), \pi_2(q))$	one word is a one-line paragraph; each further word starts a new line
6	$[\text{wrap wrap}, \text{glue}]$	$\text{F}(\text{Para}) \rightarrow \text{Para}$	$[\text{wrap wrap}, \text{glue}](\text{l}(a)) = [[a]]$, $[\text{wrap wrap}, \text{glue}](\text{r}(q)) = \text{glue}(\pi_1(q), \pi_2(q))$	one word is a one-line paragraph; each further word joins the first line
7	$[\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}]$	$\text{F}(\text{Para}) \rightarrow \text{Para}$	$[\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}] \triangleq [\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}]$	each further word either starts a new line or joins the first line
8	partition	$\text{L}(\text{Word}) \rightarrow \text{Para}$	$\text{partition} \triangleq ([\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}])$, $\text{partition}([a], \text{yss}) \iff \text{yss} = [[a]]$, $\text{partition}([a] \# \text{xs}, \text{yss}) \iff (\exists \text{xss}. \text{partition}(\text{xs}, \text{xss}) \wedge (\text{yss} = \text{new}(a, \text{xss}) \vee \text{yss} = \text{glue}(a, \text{xss})))$	every way of breaking the words into lines
9	width	$\text{Line} \rightarrow \text{Int}$	$\text{width}[a] = \text{len}(a)$, $\text{width}([a] \# \text{xs}) = (\text{len}(a) + \text{width}(\text{xs})) + 1$	the lengths of the words plus one space between neighbours
10	head	$\text{Para} \rightarrow \text{Line}$	$\text{head}[\text{xs}] = \text{xs}$, $\text{head}([\text{xs}] \# a) = \text{xs}$	the first line of the paragraph
11	$\text{fits}(w)$	$\text{P}(\text{Para})$	$\text{fits}(w, [\text{xs}]) = \text{width}(\text{xs}) \leq w$, $\text{fits}(w, [\text{xs}] \# a) = \text{width}(\text{xs}) \leq w \wedge \text{fits}(w, a)$	every line is at most w wide
12	$\text{fits}(w)$	$\text{Para} \rightarrow \text{Para}$	$\text{fits}(w)(\text{xss}, \text{yss}) \iff \text{xss} = \text{yss} \wedge \text{fits}(w, \text{xss})$	the paragraphs whose every line is at most w wide
13	$\text{ok}(w)$	$\text{Para} \rightarrow \text{Para}$	$\text{ok}(w)(\text{xss}, \text{yss}) \iff \text{xss} = \text{yss} \wedge \text{width}(\text{head}(\text{xss})) \leq w$	the paragraphs whose first line is at most w wide
14	$\text{sqr}(n)$	Int	$\text{sqr}(n) \triangleq nn$	n squared

	name	type	definition	meaning
15	<code>waste(w)</code>	<code>Para→Int</code>	$\text{waste}(w, [xs]) = 0, \text{waste}(w, [xs]\#a) = \text{sqr}(w - \text{width } xs) + \text{waste}(w, a)$	the squared white space left on every line but the last, added up
16	<code>R</code>	<code>Para→Para</code>	$R = \text{waste } w \leq \text{waste } w^\circ, R(xss, yss) \iff \text{waste}(w, xss) \leq \text{waste}(w, yss)$	p wastes no more than q
17	<code>Q(len, w)</code>	<code>Para→Para</code>	$Q(\text{len}, w) = R \cap (\text{head } \text{head}^\circ), Q(\text{len}, w)(xss, yss) \iff \text{waste}(w, xss) \leq \text{waste}(w, yss) \wedge \text{head}(xss) = \text{head}(yss)$	p wastes no more than q and has the same first line
18	<code>start(a)</code>	<code>[Para]</code>	$\text{start}(a) \triangleq [[a]]$	the one candidate for a single word: the paragraph of one line holding that word

(8.5b)

	the law	what it says
1	$(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$	glue is not monotonic on R : the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this
2	$F(Q(\text{len}, w)) \text{ [wrap wrap, new]} \sqsubseteq \text{[wrap wrap, new]} Q(\text{len}, w)$ $(\forall a. 0 \leq \text{len}(a)) \implies F(Q(\text{len}, w)) \text{ [wrap wrap, glue]} \text{ ok}(w) \sqsubseteq \text{[wrap wrap, glue]} \text{ ok}(w) Q(\text{len}, w)$ $\text{cons monotonic on collect} \leq \text{collect}^\circ.$	both halves are monotonic on Q once ties in waste are broken by the first line
3	$\text{merge}(\tau) = \text{cat}$	τ needs no sorting at all
4	$F(\tau) \text{ [wrap wrap, new]} \sqsubseteq \text{[wrap wrap, new]} \tau$ $F(\tau) \text{ [wrap wrap, glue]} \sqsubseteq \text{[wrap wrap, glue]} \tau$	both algebras are monotonic on τ , so τ is the sort order P

§8.5.1. Derivation

§8.6. Bitonic tours

(8.6a)

	name	type	definition	meaning
1	Journey(City)	Type	$\text{Journey}(\text{City}) \triangleq \mu F$	a list of at least two cities, the last two held as a pair
2	Tour(City)	Type	$\text{Tour}(\text{City}) \triangleq \text{Journey}(\text{City}) \times \text{Journey}(\text{City})$	the outward journey and the return journey
3	head	$\text{Journey}(\text{City}) \rightarrow \text{City}$	$\text{head}(\text{wrap}(p)) = \pi_1(p), \text{head}([a]\#a') = a$	the first city of the journey
4	next	$\text{Journey}(\text{City}) \rightarrow \text{City}$	$\text{next}(\text{wrap}(p)) = \pi_2(p), \text{next}([a]\#a') = \text{head}(a')$	the second city of the journey
5	$[a]\#\text{tail}(\cdot)$	$\text{Journey}(\text{City}) \rightarrow \text{Journey}(\text{City})$	$[a]\#\text{tail}(\text{wrap}(p)) = \text{wrap}(a, \pi_2(p)), [a]\#\text{tail}([a']\#a'') = [a]\#a''$	replace the first city of the journey by a
6	outcost	$\text{Journey}(\text{City}) \rightarrow \text{Int}$	$\text{outcost}(\text{wrap}(p)) = \text{tc}(p), \text{outcost}([a]\#a') = \text{tc}(a, \text{head}(a')) + \text{outcost}(a')$	the cost of travelling the journey forwards
7	incost	$\text{Journey}(\text{City}) \rightarrow \text{Int}$	$\text{incost}(\text{wrap}(p)) = \text{tc}(\pi_2(p), \pi_1(p)), \text{incost}([a]\#a') = \text{tc}(\text{head}(a'), a) + \text{incost}(a')$	the cost of travelling the journey backwards
8	cost(t)	Int	$\text{cost}(t) \triangleq \text{outcost}(\pi_1(t)) + \text{incost}(\pi_2(t))$	the outward journey forwards plus the return journey backwards
9	start	$\text{City} \times \text{City} \rightarrow \text{Tour}(\text{City})$	$\text{start}(a) \triangleq (\text{wrap}(a), \text{wrap}(a))$	the tour of the last two cities, both journeys being that one edge
10	dropl(a,t)	Tour(City)	$\text{dropl}(a,t) \triangleq ([a]\#\text{tail}(\pi_1(t)), [a]\#\pi_2(t))$	a replaces the first city of the outward journey and is put in front of the return
11	dropr(a,t)	Tour(City)	$\text{dropr}(a,t) \triangleq ([a]\#\pi_1(t), [a]\#\text{tail}(\pi_2(t)))$	a is put in front of the outward journey and replaces the first city of the return
12	$[\text{start}, \text{dropl}]$	$F(\text{Tour}(\text{City})) \rightarrow \text{Tour}(\text{City})$	$[\text{start}, \text{dropl}](l(p)) = \text{start}(p), [\text{start}, \text{dropl}](r(q)) = \text{dropl}(\pi_1(q), \pi_2(q))$	start at the last two cities, then add each city by dropl

	name	type	definition	meaning
13	$[\text{start}, \text{dropr}]$	$F(\text{Tour}(\text{City})) \rightarrow \text{Tour}(\text{City})$	$[\text{start}, \text{dropr}](l(p)) = \text{start}(p),$ $[\text{start}, \text{dropr}](r(q)) = \text{dropr}(\pi_1(q), \pi_2(q))$	start at the last two cities, then add each city by dropr
14	$[\text{start}, \text{dropl}] \cup [\text{start}, \text{dropr}]$	$F(\text{Tour}(\text{City})) \rightarrow \text{Tour}(\text{City})$	$[\text{start}, \text{dropl}] \cup [\text{start}, \text{dropr}] \triangleq$ $[\text{start}, \text{dropl}] \cup [\text{start}, \text{dropr}],$ $([\text{start}, \text{dropl}] \cup [\text{start}, \text{dropr}])(u, t) = t =$ $[\text{start}, \text{dropl}](u) \vee t = [\text{start}, \text{dropr}](u)$	each further city is added by dropl or by dropr
15	$\langle [\text{start}, \text{dropl}] \cup [\text{start}, \text{dropr}] \rangle$	$\mu F \rightarrow \text{Tour}(\text{City})$	$\langle [\text{start}, \text{dropl}] \cup [\text{start}, \text{dropr}] \rangle \triangleq$ $\langle [\text{start}, \text{dropl}] \cup [\text{start}, \text{dropr}] \rangle$	every bitonic tour of the cities
16	R	$\text{Tour}(\text{City}) \rightarrow \text{Tour}(\text{City})$	$R = \text{cost} \leq \text{cost}^\circ, R(t, t') \iff \text{cost}(t) \leq \text{cost}(t')$	t costs no more than t'
17	$\text{next2}(t)$	$\text{City} \times \text{City}$	$\text{next2}(t) \triangleq (\text{next}(\pi_1(t)), \text{next}(\pi_2(t)))$	the second cities of both journeys
18	$\text{head2}(t)$	$\text{City} \times \text{City}$	$\text{head2}(t) \triangleq (\text{head}(\pi_1(t)), \text{head}(\pi_2(t)))$	the first cities of both journeys
19	Q	$\text{Tour}(\text{City}) \rightarrow \text{Tour}(\text{City})$	$Q = R \cap (\text{next2 next2}^\circ) \cap (\text{head2 head2}^\circ),$ $Q(t, t') \iff \text{cost}(t) \leq \text{cost}(t') \wedge$ $(\text{next}(\pi_1(t)) = \text{next}(\pi_1(t')) \wedge \text{next}(\pi_2(t)) = \text{next}(\pi_2(t'))) \wedge$ $\text{head}(\pi_1(t)) = \text{head}(\pi_1(t')) \wedge \text{head}(\pi_2(t)) = \text{head}(\pi_2(t'))$	t costs no more than t' , and both journeys of t and t' share their first two cities

(8.6b)

	the law	what it says
1	$(1 \times R) \text{ dropl} \sqsubseteq \text{dropl} \quad R \quad \text{FALSE}$ $(1 \times R) \text{ dropr} \sqsubseteq \text{dropr} \quad R \quad \text{FALSE}$	neither drop is monotonic on R : the two edges it adds and removes depend on head and next of both lists
2	$F(Q) \text{ } [\text{start}, \text{dropl}] \sqsubseteq [\text{start}, \text{dropl}] \quad Q$ $F(Q) \text{ } [\text{start}, \text{dropr}] \sqsubseteq [\text{start}, \text{dropr}] \quad Q,$	both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input
3	$F(T) \text{ } [\text{start}, \text{dropl}] \sqsubseteq [\text{start}, \text{dropl}] \quad T$ $F(T) \text{ } [\text{start}, \text{dropr}] \sqsubseteq [\text{start}, \text{dropr}] \quad T$	both algebras are monotonic on T , so T is the sort order P

§8.6.1. Derivation

$(\text{listcp } \langle g_1, g_2 \rangle \text{ cat thinlist}(Q)) \text{ minlist}(R) \sqsubseteq \frac{(\text{list}(\text{start}, \text{dropl}) \cup \text{list}(\text{start}, \text{dropr}))}{\exists} \text{ est}(R) \quad (8.6.1a)$ a least-cost bitonic tour, as a fold that thins the tours kept at each city	
circuit	Hinze–Marsden
$\mu F \xrightarrow{\frac{1}{\exists}} P(\mu F) \xrightarrow{E((\text{list}(\text{start}, \text{dropl}) \cup \text{list}(\text{start}, \text{dropr})))} P(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City})$ $\frac{(\text{list}(\text{start}, \text{dropl}) \cup \text{list}(\text{start}, \text{dropr}))}{\exists} \text{ est}(R)$	
$\mu F \xrightarrow{\frac{1}{\exists}} P(\mu F) \xrightarrow{E((\text{list}(\text{start}, \text{dropl}) \cup \text{list}(\text{start}, \text{dropr})))} P(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City})$ $= \frac{(\text{list}(\text{start}, \text{dropl}) \cup \text{list}(\text{start}, \text{dropr}))}{\exists} \text{ est}(R)$ $(\text{list}(\text{start}, \text{dropl}) \cup \text{list}(\text{start}, \text{dropr})) \triangleq (\text{list}(\text{start}, \text{dropl}) \cup \text{list}(\text{start}, \text{dropr}))$	
$\mu F \xrightarrow{\left[\frac{F(\text{list}(\text{Journey}(\text{City})^2))}{\text{listcp}} \text{ list}(F(\text{Journey}(\text{City})^2)) \right]} \left[\frac{g_1}{g_2} \right] \text{ cat thinlist}(Q) \text{ minlist}(R)$ binary thinning theorem, at $P \triangleq T$ with merge $T = \text{cat}$, Q from $F(Q)$ $[\text{start}, \text{dropl}] \sqsubseteq [\text{start}, \text{dropl}] Q, F(Q) [\text{start}, \text{dropr}] \sqsubseteq [\text{start}, \text{dropr}] Q$	
$(\text{list}(\text{wrap}, \text{cpr } \langle \text{list}(\text{dropl}), \text{list}(\text{dropr}) \rangle \text{ cat thinlist}(Q)) \text{ minlist}(R)$ $\text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{start}), \text{list}(\text{drop}_i)] \text{ —}$ $= [\text{start}, \text{dropl}](l(p)) = \text{start}(p), [\text{start}, \text{dropl}](r(q)) = \text{dropl}(\pi_1(q), \pi_2(q)), [\text{start}, \text{dropr}](l(p)) = \text{start}(p),$ $[\text{start}, \text{dropr}](r(q)) = \text{dropr}(\pi_1(q), \pi_2(q)); \text{quadratic, two tours added per step}$	